Rutgers Health Hack Challenge 1

Context-Aware Hospital Workflow Guidance for Tele-Neurology

Sevaro | Challenge Brief + Synthetic Hospital Guide Dataset | v2

The Challenge

Sevaro clinicians provide tele-neurology services across more than 200 hospitals. Providers use Synapse for consultation workflows, documentation, charting, video and phone consults, and clinical decision support. Each hospital has its own workflow configuration: where the video evaluation happens, which thrombolytic is used and who mixes it, whether the teleneurologist may enter orders, how a large vessel occlusion is escalated, which note title is required, and whether a charge must be entered before sign-off. A real hospital guide has more than 80 of these fields.

Design a workflow companion that puts the right hospital-specific rule in front of the provider at the moment they need it, inside the caselog, without making them leave the encounter to search a guide.

Why This Matters

Today a provider who needs a site rule has to pause documentation, open the guide, find the hospital and section, then return to the caselog. During an acute stroke evaluation that interruption competes with time-critical clinical decisions, so providers often rely on memory instead. Memory is frequently wrong for a given site.

The errors that result (wrong order set, wrong note title, missed charge entry, wrong escalation path) are usually caught only after the case closes. Each one then needs follow-up between the provider and a quality team, and sometimes the hospital. **The goal is to prevent those errors during the encounter, not just to make the guide easier to search.**

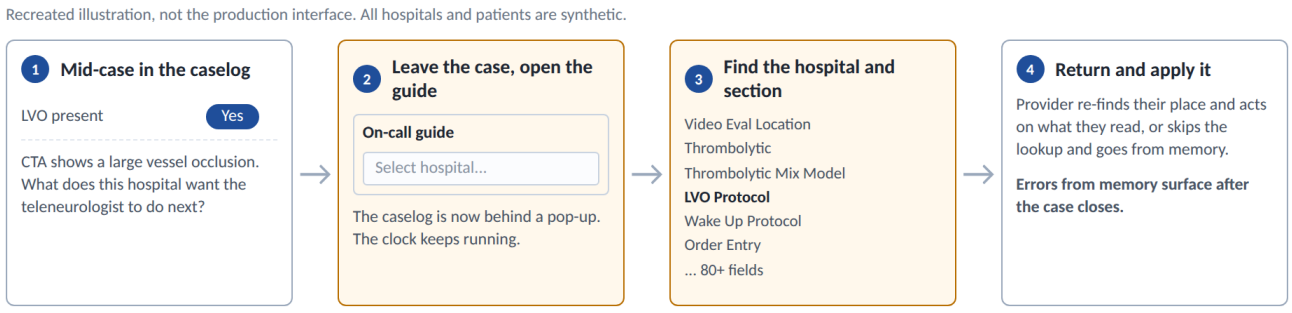


Figure 1. Today's lookup loop. Your nudges should remove steps 2 to 4.

Primary deliverable: proactive nudges. The system watches what the provider is doing in the caselog and surfaces the relevant rule at the right field, unprompted. **Secondary deliverable: on-demand answers** to free-text questions. A chat assistant on its own, without proactive nudges, will not place highly.

Inputs Your System Receives

Your system is given an **encounter context** every time the provider moves to a new caselog step or changes a field. "Context-aware" means using all of it, not just the hospital ID.

Field	Example	Why it matters
hospital_id	SYN-006	Which guide applies. Answers must never come from another hospital.
consult_type / encounter_type	Stroke / Acute	Some rules only apply to stroke alerts or to rounding
caselog_step	Imaging	Where in the workflow the provider is
field_event	lvo_present = true	The specific action that may trigger a nudge
local_time + local_timezone	19:30 America/Chicago	Many rules depend on day and time at the hospital
provider_timezone	America/New_York	Providers often work in a different time zone from the hospital

Field	Example	Why it matters
charge_entered	false	Encounter state needed for sign-off checks

Caselog Steps and Nudge Triggers

A simplified caselog model. Each row is a point where a nudge may be useful. You may add, remove or re-tier triggers, but explain why.

#	Step	Caselog field	Trigger	Guide fields	Default tier
1	Intake	Reason for consult / hospital selected	Encounter opened for a hospital	Video Eval Location; Wake Up Protocol	Info
2	History	Last known well = Unknown	Toggle set to Unknown	Wake Up Protocol	Advisory
3	History	Allergies	Thrombolytic allergy entered	Thrombolytic	Advisory
4	Imaging	CTA done = Yes	CTA result being entered	CT Perfusion; AI	Info
5	Imaging	Large vessel occlusion present = Yes	Set to Yes	LVO Protocol (evaluate against local time)	Advisory
6	Recommendation	Thrombolytic given = Yes	Set to Yes	Thrombolytic; Thrombolytic Mix Model; Order Entry; Order Set Name	Advisory
7	Recommendation	Thrombolytic order placed in EHR?	Field displayed	Order Entry; Order Set Name	Advisory
8	Recommendation	Assessment / plan	Seizure or EEG mentioned	EEG (evaluate against local time)	Info
9	Note	Note title	Note being created	Required Note Title	Advisory
10	Sign-off	Complete consult	Provider clicks Complete	Charge Entry; Consult Completion Rule	Hard stop
11	Rounding	Rounding encounter opened	Encounter type = Rounding	Rounding Timing; Rounding Orders	Info

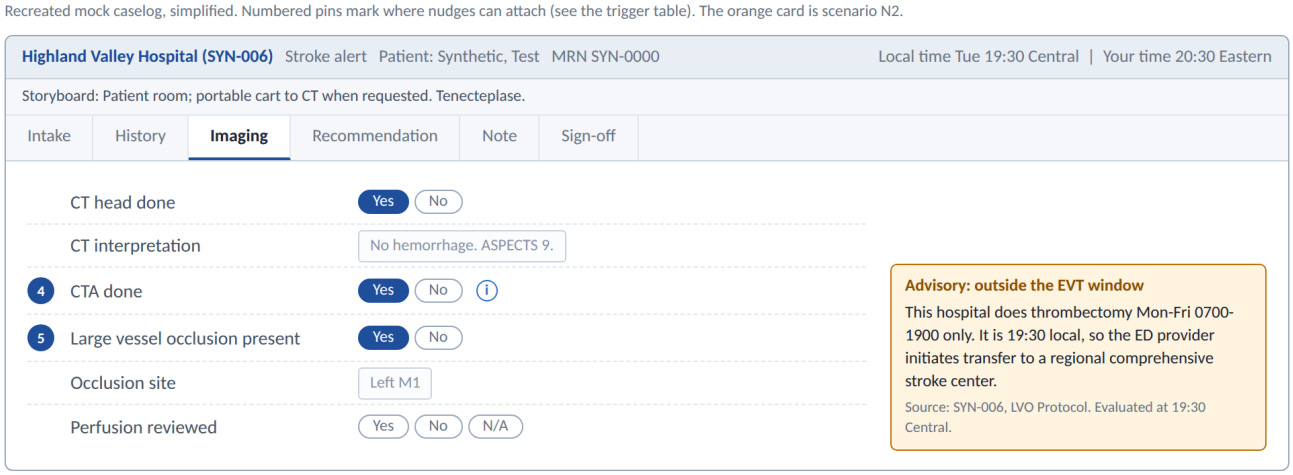


Figure 2. Recreated mock of the caselog Imaging step. Pins 4 and 5 match rows 4 and 5 of the table above. The orange card is what scenario N2 should produce. You may build your own caselog mock; it does not need to look like this.

Nudge tiers

Tier	Behavior	Example
Hard stop	Blocks the action until the provider acknowledges or fixes it	Charge must be entered before sign-off
Advisory	Visible next to the field; does not block	This hospital is not thrombectomy capable; ED provider starts the transfer
Info	Available on hover or in a side panel	CT perfusion is not available here

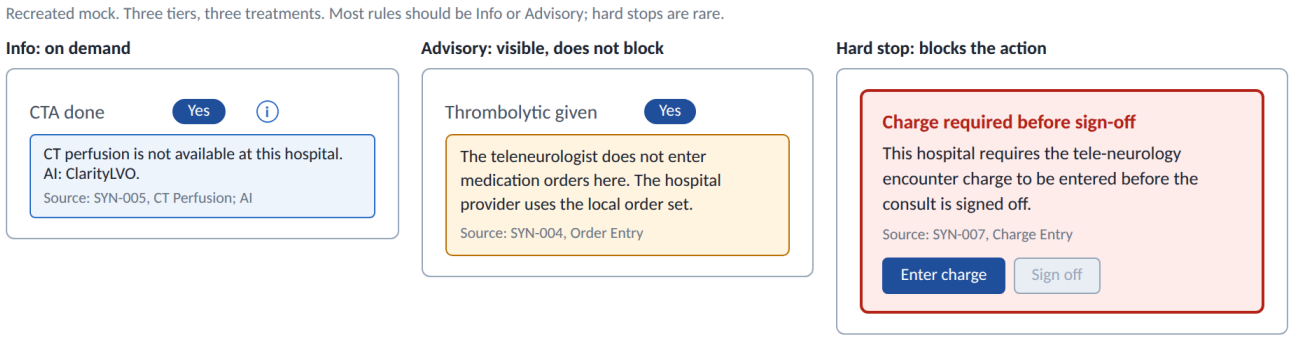


Figure 3. One way to show the three tiers. Treatments are illustrative; the behavior is what matters.

Nudge fatigue is the main design risk. With 80+ fields per hospital, showing everything trains providers to ignore all of it. Show that your system stays quiet when nothing applies, and justify each nudge that interrupts.

Required Answer Behavior

Every answer and every nudge must cite the hospital and the guide field it came from, and must carry one of these statuses:

Status	When	What the provider should see
Answered	The field exists and is current	The value, with source field
Not in guide	The field is missing for this hospital	"Not available in the hospital guide." Never fill it from another hospital.
Time-dependent	The answer depends on the local day or time	The answer for now, and the rule used to decide it
Stale	The field was last verified more than 180 days before the encounter	The value, clearly flagged as possibly out of date
Conflict	The guide and the hospital storyboard disagree	Both values with their sources; never silently pick one
Hard stop	A required step is incomplete	What is missing and what to do

If a time-dependent rule refers to hours the guide does not define (for example, "staffed hours" with no hours given), say so. Do not assume a window.

Non-Functional Requirements

- Nudges appear within 500 ms of the triggering field change. Load the hospital's guide when the encounter opens, not per field.
- If guide data is unavailable, the caselog keeps working and the provider is told guidance is unavailable.
- Read-only: nothing the provider does changes the guide.
- **Adding a hospital or a field is a data change, not a code change.** Judges will add a new hospital during your demo.

Data Package

File	Contents
hospital_guides.json	All seven guides, one record per field with value and last-verified date
storyboard_notes.json	Second-source values that may agree or conflict with the guide
caselog_field_map.csv	Caselog steps, fields, triggers and mapped guide fields
encounter_context_example.json	Example of the context object your system receives
validation_set.csv	V01-V20 questions with expected status and answer
nudge_scenarios.json	N1-N9 encounter scenarios with expected nudges

Synthetic Hospital Guides

Seven fictional hospitals with intentionally different configurations

All fields were last verified on 2026-02-02 unless marked otherwise. The reference encounter date for staleness checks is 2026-03-17. Some fields are deliberately missing for some hospitals.

Cedar Ridge Medical Center (SYN-001)

Guide Field	Synthetic Value
Local Time Zone	Central (America/Chicago)
Video Eval Location	Emergency department patient room
Thrombolytic	Alteplase
Thrombolytic Mix Model	Pharmacy prepares and delivers to bedside
LVO Protocol	Thrombectomy capable during staffed neurointerventional coverage. Teleneurologist alerts ED provider and activates the synthetic neurovascular response line. Outside coverage hours, use the transfer pathway.
Wake Up Protocol	Yes
Order Entry	Thrombolytic order set and post-treatment monitoring orders only
Order Set Name	Search "Acute Stroke Remote" - "CRMC Acute Ischemic Stroke - Alteplase [SYN-41021]"
Charge Entry	No
Required Note Title	Tele-Neurology Consult
Rounding Process	Hospitalist service sends daily census to the rounding queue by 0630 Central Time
When Do We Round?	Monday-Friday
Rounding Timing	0700-1100 Central Time
Rounding Orders	No routine orders
General Neurology	Local group covers outpatient; Sevaro covers inpatient tele-neurology
Pediatric Neurology	Regional children's center via transfer line
EEG	Mon-Sat 0700-1900
CT Head	24/7
CTA Head/Neck	24/7
CT Perfusion	24/7
MRI	Daily 0600-2300; overnight on-call
AI	RapidView Neuro - synthetic
EHR	MedAxis - synthetic
PACS	NovaPACS - synthetic
VPN	SecureBridge - synthetic

Bluewater Regional Hospital (SYN-002)

Guide Field	Synthetic Value
Local Time Zone	Mountain (America/Denver)
Video Eval Location	CT holding bay for stroke alerts; patient room for other consults
Thrombolytic	Tenecteplase
Thrombolytic Mix Model	ED nurse mixes from stroke kit after verification
LVO Protocol	Not thrombectomy capable. Teleneurologist communicates suspected LVO to ED provider. ED provider activates transfer center. Synthetic preferred destination: Metro Neuroscience Institute.
Wake Up Protocol	No
Order Entry	Thrombolytic order entry only
Order Set Name	Search "Neuro TNK" - "BWRH Stroke - Tenecteplase [SYN-88210]"
Charge Entry	Yes
Rounding Process	Charge nurse initiates rounding request in Synapse
When Do We Round?	Daily
Rounding Timing	0830-1200 Mountain Time
Rounding Orders	No
Cardiovascular Services	Mon-Fri 0700-1800; weekend on-call
Carotid Ultrasound	Daily 0800-1700
Cardiac Monitoring	Inpatient telemetry 24/7; outpatient monitor via cardiology clinic
EEG	Mon-Fri 0700-1700
TEE	Mon-Fri 0800-1600
PT/OT/ST	PT daily; OT Mon-Sat; ST Mon-Fri
Specialty Labs	ReferenceLab North - synthetic
EHR	CareSphere - synthetic
PACS	ImageHarbor - synthetic

Redwood University Medical Center (SYN-003)

Guide Field	Synthetic Value
Local Time Zone	Pacific (America/Los_Angeles)
Video Eval Location	Patient room
Thrombolytic	Tenecteplase
Thrombolytic Mix Model	Pharmacy verifies; ED nurse mixes and administers
LVO Protocol	Thrombectomy capable 24/7. Teleneurologist communicates LVO concern to ED provider and activates the synthetic Neurointerventional Hotline.
Wake Up Protocol	Yes
Order Entry	Full acute stroke order set permitted
Order Set Name	Search "RUMC Stroke Pathway" - "Acute Stroke / TNK / EVT Evaluation [SYN-73004]"
Charge Entry	No
Required Note Title	Neurology Consult - Remote
Rounding Process	Scheduled inpatient list appears automatically in Synapse
When Do We Round?	Daily
Rounding Timing	0600-1400 Pacific Time
Rounding Orders	Yes - inpatient neurology orders permitted
General Neurology	Sevaro inpatient; university faculty outpatient
Pediatric Neurology	In-house pediatric neurology 24/7
EEG	Routine and continuous EEG 24/7
CT Head	24/7
CTA Head/Neck	24/7
CT Perfusion	24/7
MRI	24/7
AI	NeuroSignal AI - synthetic
EHR	Epic synthetic training tenant
EHR Department Login Location	RUMC TELE NEUROLOGY
PACS	VisionGrid - synthetic
VPN	CloudGate Health - synthetic

Meadowlands Community Hospital (SYN-004)

Guide Field	Synthetic Value
Local Time Zone	Eastern (America/New_York)
Video Eval Location	Emergency department patient room
Thrombolytic	Alteplase
Thrombolytic Mix Model	Pharmacy mixes; courier delivers to ED
LVO Protocol	Not thrombectomy capable. Teleneurologist communicates LVO concern to primary provider. Primary provider contacts the synthetic Regional Transfer Hub; destination depends on bed availability.
Wake Up Protocol	No
Order Entry	No medication orders entered by teleneurologist
Order Set Name	Hospital provider uses local stroke order set
Charge Entry	No
Required Note Title	Neuro Tele Consult
Consult Completion Rule	Do not complete the consult in Synapse until the hospital provider confirms the patient plan
Rounding Process	House supervisor calls Sevaro operations to initiate rounding
When Do We Round?	Tuesday, Thursday, Saturday
Rounding Timing	0730-1030 Eastern Time
Rounding Orders	No
General Neurology	Sevaro is the only inpatient neurology service
Pediatric Neurology	Transfer to regional pediatric center
Cardiovascular Services	Mon-Fri 0800-1700
Carotid Ultrasound	Mon-Fri 0700-1600
Cardiac Monitoring	Inpatient telemetry; outpatient monitoring arranged after discharge
EEG	Tue/Thu 0800-1500 **[last verified 2025-06-10]**
TEE	Tue-Fri 0800-1500
Stress Testing	Mon-Fri 0700-1400
PT/OT/ST	PT daily; OT Mon-Fri; ST Mon/Wed/Fri
Specialty Labs	Apex Reference Labs - synthetic

Suncrest Medical Pavilion (SYN-005)

Guide Field	Synthetic Value
Local Time Zone	Eastern (America/New_York)
Video Eval Location	CT room for stroke activation; ICU or patient room for inpatient consults
Thrombolytic	Tenecteplase
Thrombolytic Mix Model	Pharmacy prepares after teleneurology recommendation
LVO Protocol	Not thrombectomy capable. Teleneurologist informs ED provider and then calls the synthetic Transfer Command Center. Synthetic preferred destination: Capital Neurovascular Center.
Wake Up Protocol	Yes
Order Entry	Thrombolytic and imaging orders permitted
Order Set Name	Search "Stroke TeleCare" - "Suncrest Acute Stroke - TNK [SYN-55190]"
Charge Entry	Yes
Rounding Process	Hospitalist coordinator schedules rounding in Synapse one day in advance
When Do We Round?	As scheduled
Rounding Timing	0700-1700 Eastern Time
Rounding Orders	Medication changes require hospitalist co-sign
VPN	NorthPass - synthetic
EHR	MedConnect One - synthetic
EHR Department Login Location	SCMP REMOTE NEURO
CT Head	24/7
CTA Head/Neck	24/7
CT Perfusion	No
AI	ClarityLVO - synthetic
MRI	Mon-Fri 0500-2300; weekend 0700-1900; overnight call
PACS	ClearImage Enterprise - synthetic
EEG	Mon-Fri 0800-1800; weekend on-call

Highland Valley Hospital (SYN-006)

Guide Field	Synthetic Value
Local Time Zone	Central (America/Chicago)
Video Eval Location	Patient room; portable cart to CT when requested
Thrombolytic	Tenecteplase
Thrombolytic Mix Model	ED pharmacist prepares during staffed hours; nursing mixes overnight **[last verified 2025-08-15]**
LVO Protocol	Thrombectomy capable Mon-Fri 0700-1900. During coverage, teleneurologist activates the synthetic EVT team. Outside coverage, ED provider initiates transfer to a regional comprehensive stroke center.
Wake Up Protocol	Yes
Order Entry	Thrombolytic order set only
Order Set Name	Search "HVH Telestroke" - "Acute Ischemic Stroke - Tenecteplase [SYN-66018]"
Charge Entry	No
Required Note Title	MD Progress Note - Neuro
Rounding Process	Unit clerk sends request through hospital workflow queue
When Do We Round?	Monday-Saturday
Rounding Timing	0800-1500 Central Time
Rounding Orders	No
General Neurology	Local neurologist until 1700; Sevaro nights and overflow
Pediatric Neurology	Children's specialty network; phone consultation first
Cardiovascular Services	Daily 0700-1900
Carotid Ultrasound	Mon-Sat 0700-1800
Cardiac Monitoring	Telemetry 24/7; outpatient patch monitor
EEG	Daily 0700-1900; continuous EEG unavailable
TEE	Mon-Fri 0700-1700
Stress Testing	Mon-Sat 0700-1500
PT/OT/ST	PT/OT daily; ST Mon-Sat
Specialty Labs	Meridian Diagnostics - synthetic

Silver Oak Health Center (SYN-007)

Guide Field	Synthetic Value
Local Time Zone	Eastern (America/New_York)
Video Eval Location	Patient room
Thrombolytic	Alteplase
Thrombolytic Mix Model	ED nursing prepares from pharmacy-issued kit
LVO Protocol	Not thrombectomy capable. Teleneurologist communicates LVO concern to ED provider; ED provider calls synthetic Statewide Transfer Access.
Wake Up Protocol	No
Order Entry	No orders entered by teleneurologist
Order Set Name	Not applicable
Charge Entry	Yes - tele-neurology encounter charge before sign-off
Required Note Title	Tele-Neurology Encounter
Rounding Process	Hospitalist sends secure message to initiate rounding
When Do We Round?	Monday-Friday
Rounding Timing	0900-1300 Eastern Time
Rounding Orders	No
VPN	EdgeConnect Health - synthetic
EHR	ClinixPro - synthetic
EHR Department Login Location	SOHC NEUROLOGY REMOTE
CT Head	24/7
CTA Head/Neck	24/7
CT Perfusion	Mon-Fri 0700-1900
AI	None
MRI	Daily 0700-2200; no overnight coverage
PACS	RadiantView - synthetic
EEG	Mon-Fri 0900-1600

Hospital Storyboard Notes (Second Source)

Synapse also shows a short hospital storyboard in the caselog header. It is maintained separately and can disagree with the guide.

Hospital	Field	Storyboard value	Updated	Note
SYN-002	Video Eval Location	Patient room	2025-09-14	Hospital storyboard card shown in the caselog header. Disagrees with the on-call guide for stroke alerts.
SYN-005	Thrombolytic	Tenecteplase	2026-01-20	Agrees with the guide. Included so teams do not flag every second source as a conflict.

Prototype Validation Examples

Questions to test that the solution uses the correct hospital context. Encounter times are local to the hospital unless stated.

#	Hospital	Provider question	Local time	Status	Expected grounded response
V01	SYN-001	Where should I perform the video evaluation?	Tue 2026-03-17 14:00	answered	Emergency department patient room.
V02	SYN-002	What is the LVO workflow?	Tue 2026-03-17 14:00	answered	ED provider activates transfer center; synthetic preferred destination is Metro Neuroscience Institute.
V03	SYN-003	Can I place inpatient neurology orders during rounding?	Tue 2026-03-17 09:00	answered	Yes, according to this synthetic hospital configuration.
V04	SYN-004	Is EEG available on Friday?	Fri 2026-03-20 10:00	not in guide	The guide only lists Tue/Thu; Friday availability is not provided. Do not assume.

#	Hospital	Provider question	Local time	Status	Expected grounded response
V05	SYN-005	Does the hospital have CT perfusion?	Tue 2026-03-17 14:00	answered	No.
V06	SYN-006	What is the LVO pathway after 1900 on a weekday?	Tue 2026-03-17 19:30	time dependent	The staffed thrombectomy window has ended; ED provider initiates transfer.
V07	SYN-007	Which order set should I use?	Tue 2026-03-17 14:00	answered	Not applicable; the teleneurologist does not enter orders.
V08	SYN-001	Patient has an LVO. What do I do?	Tue 2026-03-17 22:30	time dependent	Pathway depends on staffed neurointerventional coverage, but the guide does not give coverage hours. State both branches and say coverage hours are not in the guide. Do not borrow hours from another hospital.
V09	SYN-006	Patient has an LVO. Do I activate the EVT team?	Sat 2026-03-21 10:00	time dependent	No. Coverage is Mon-Fri only. ED provider initiates transfer to a regional comprehensive stroke center.
V10	SYN-007	Can I sign off now?	Tue 2026-03-17 14:00	hard stop	Not until the tele-neurology encounter charge is entered.
V11	SYN-005	Can I change the antiplatelet during rounding?	Tue 2026-03-17 10:00	answered	Medication changes require hospitalist co-sign.
V12	SYN-002	Where do I do the video eval for a stroke alert?	Tue 2026-03-17 14:00	conflict	On-call guide: CT holding bay for stroke alerts. Storyboard: patient room. Show both with sources; do not silently pick one.
V13	SYN-004	Is EEG available on Thursday?	Thu 2026-03-19 10:00	stale	Guide lists Tue/Thu 0800-1500, so yes, but this field was last verified 2025-06-10. Flag it as possibly out of date.
V14	SYN-002	Who covers pediatric neurology?	Tue 2026-03-17 14:00	not in guide	Not available in this hospital's guide. Do not use another hospital's answer.
V15	SYN-005	Which thrombolytic order set do I search for?	Tue 2026-03-17 14:00	answered	Search "Stroke TeleCare" - "Suncrest Acute Stroke - TNK [SYN-55190]".
V16	SYN-007	Can I get CT perfusion on Saturday?	Sat 2026-03-21 11:00	time dependent	No. CT perfusion is Mon-Fri 0700-1900 only.
V17	SYN-002	Is the rounding window open? (Provider is in Eastern time.)	Tue 2026-03-17 08:15 local (10:15 ET)	time dependent	Not yet. Window is 0830-1200 Mountain; it opens in 15 minutes (10:30 ET).
V18	SYN-006	Who mixes the TNK?	Wed 2026-03-18 20:00	time dependent	Guide says pharmacist during staffed hours, nursing overnight, but staffed pharmacy hours are not defined. Field last verified 2025-08-15, so also flag it as stale. Say what is unknown; do not guess the cutoff.
V19	SYN-003	What should I title the note?	Tue 2026-03-17 14:00	answered	Neurology Consult - Remote.
V20	SYN-005	What note title is required?	Tue 2026-03-17 14:00	not in guide	Not available in this hospital's guide.

Nudge Trigger Scenarios

Encounter situations and the nudge a good solution should produce. Scenario N8 checks that the system stays quiet.

#	Hospital	Type	Step	Field event	Local time	Expected nudge
N1	SYN-002	Stroke	Imaging	LVO present = Yes	Tue 14:00 Mountain	Advisory: not thrombectomy capable. Tell the ED provider; they activate the transfer center. Preferred destination: Metro Neuroscience Institute.
N2	SYN-006	Stroke	Imaging	LVO present = Yes	Tue 19:30 Central	Advisory: outside the EVT window (Mon-Fri 0700-1900). ED provider initiates transfer. Must not suggest activating the EVT team.
N3	SYN-003	Stroke	History	Last known well = Unknown	Tue 03:10 Pacific	Advisory: wake-up protocol is in use at this hospital.
N4	SYN-004	Stroke	Recommendation	Thrombolytic given = Yes	Tue 16:00 Eastern	Advisory: teleneurologist does not enter medication orders; hospital provider uses the local order set. No order-set name nudge.
N5	SYN-007	Stroke	Sign-off	Complete clicked; charge not entered	Tue 14:00 Eastern	Hard stop: tele-neurology encounter charge required before sign-off.
N6	SYN-005	Stroke	Imaging	CTA done = Yes	Tue 14:00 Eastern	Info: CT perfusion not available; AI is ClarityLVO.

#	Hospital	Type	Step	Field event	Local time	Expected nudge
N7	SYN-004	General Neuro	Sign-off	Complete clicked	Tue 15:00 Eastern	Hard stop: do not complete until the hospital provider confirms the plan.
N8	SYN-002	General Neuro	History	Allergies entered: none	Tue 14:00 Mountain	No nudge. Nothing in the guide applies; staying quiet is the correct behavior.
N9	SYN-001	Rounding	Rounding	Encounter opened	Sun 08:00 Central	Info: this hospital rounds Monday-Friday only; check why a Sunday rounding encounter exists.

What Teams Must Demonstrate

1. Proactive nudges driven by the encounter context, covering scenarios N1-N9, including staying quiet in N8.
2. On-demand answers that pass validation examples V01-V20, each with its source field and status.
3. Time-aware rules evaluated in the hospital's local time, including a provider in a different time zone.
4. Stale, conflicting and missing data handled visibly, never papered over.
5. **Live test.** Judges will add a new hospital guide in the same format and run unseen questions and scenarios against it.

Data and Confidentiality Boundary

No real Sevaro or hospital protocols or guides will be shared. Participants use only the synthetic hospital guides in this package. Hospital names, systems, order sets, identifiers, transfer destinations, schedules, note titles and operational rules are fictional. The dataset mirrors the kinds of fields and variability a workflow solution must handle, but does not reproduce the real hospital guides used to shape this challenge. No patient data or production access is required. This content is for hackathon prototyping only and is not clinical guidance.